Buzz | BongoHive AI Lab

Buzz
Progress report

What Buzz is for, what has been built so far, and what happens next.

BongoHive AI Lab · Data & AI team
19 October 2026

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1 · THE GOAL

What Buzz **is for**

Buzz is BongoHive's own AI phone assistant. It answers calls in natural spoken English, deals with the routine questions, and hands callers to a person when they need one.

> **In one sentence:** callers get a quick, friendly, consistent answer every time, staff spend their time on the conversations that need a person, and everything said on the call stays on BongoHive's own servers.

What Buzz will do

1 Answer calls

Picks up, tells the caller they are speaking to an AI assistant, and answers common questions: programmes, coworking plans, events, opening hours and how to find us.

2 Pass callers to staff

When someone asks for a person, or Buzz cannot help, it rings the right team. Staff hear a short introduction first and can accept or decline. If nobody is free, Buzz takes a message.

3 Make approved calls

Later on, Buzz can call people who have agreed to be contacted, for example event reminders. It only calls during permitted hours and never calls anyone who has said no.

4 Keep records tidy

After each call, a short summary goes to HubSpot, with any sensitive details such as card or ID numbers removed first.

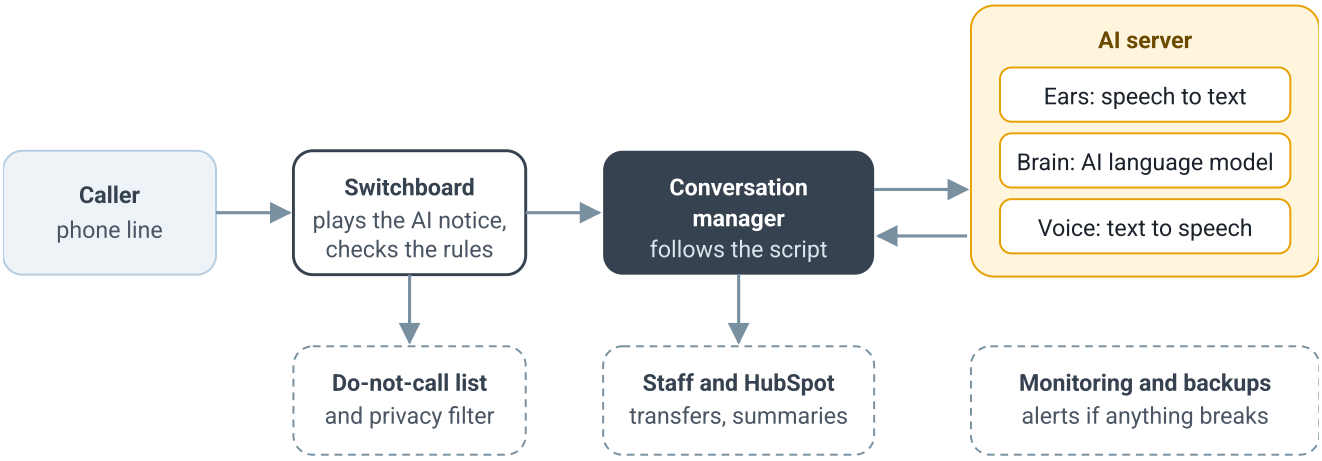
The principles it is built on

- **Private by design.** The AI that listens, thinks and speaks runs on BongoHive's own computers. No call audio, transcript or knowledge is sent to an outside AI company.
- **Honest with callers.** Every call starts with a short message saying the caller is talking to an AI assistant. It cannot be skipped or talked over.
- **Respectful of consent.** A caller can say "please don't call me again" at any time; that number is blocked straight away for all future calls.
- **Safe with personal data.** Buzz never asks for or keeps PINs, card numbers or national ID numbers. Recordings and transcripts are removed or anonymised after 90 days.

2 · HOW IT WORKS

The parts **of the system**

Buzz is made of a few parts that each do one job. They run on two BongoHive servers: one for the AI, and one for everything else.



Solid boxes are on every call. Dashed boxes support the call: checking rules, passing callers on, and keeping the system healthy.

Part	What it does, in plain terms
Switchboard (Asterisk)	Connects to the phone line. Plays the "you are speaking to an AI" notice, and refuses any outgoing call that breaks the rules before it is ever dialled.
Conversation manager (Dograh)	Runs the conversation: what Buzz says first, which questions it handles, when to transfer, and what to record afterwards. Staff manage it through a web page with the Buzz logo.
AI server (vLLM and Speaches)	Turns the caller's speech into text, decides what to say, and speaks the reply in a British English voice. All three run on our own graphics card, with no internet needed.
Do-not-call list and privacy filter	Keeps the list of people who asked not to be called, checks every outgoing number against it, and removes sensitive details from summaries before they reach HubSpot.
Monitoring and backups	A dashboard showing call volumes and response speed, alerts when something fails, nightly encrypted backups, and automatic deletion of old recordings.

3 · PROGRESS

What has **been done**

All the base code is written, set up to BongoHive's privacy rules and checked automatically. It has not yet run on real servers or a phone line, as those are not available yet.

7
areas of work completed

428
automatic checks, all passing

18
test conversations written for the front desk

Area	What was done	Status
Brand	A new Buzz logo in BongoHive AI Lab colours, and a combined Buzz and BongoHive AI Lab logo for the web page and documents.	Complete
AI server	Set-up for the speech and language models, sized for 16, 24 or 48 GB graphics cards, plus a speed test that measures how quickly Buzz replies.	Built, checked offline
Main platform	A locked-down conversation manager with Buzz branding. "Phone home" features are off and a firewall blocks anything leaving our network.	Built, checked offline
Switchboard	AI notice on every call, calling hours, the do-not-call check before every dial, screened transfers and message-taking.	Built, checked offline
Do-not-call and privacy	Do-not-call list following Zambia's numbering plan (blocks premium, short-code and international numbers), and a filter that removes card, PIN and NRC numbers.	Built, 245 checks pass
First assistant	Front-desk assistant with opt-out, transfer and end-call tools, waiting for BongoHive's information. A one-step tool connects it to the AI server.	Draft, needs content
Monitoring and backups	Dashboard in BongoHive colours, 30 alert rules, nightly encrypted backups with a restore check, and 90-day deletion of recordings and transcripts.	Built, checked offline

The 428 checks run automatically every time the code changes, and a separate checker stops anyone saving an assistant that breaks the privacy rules.

4 · WHAT WE LEARNED

Problems found **and fixed early**

Building against the real source code turned up several issues that would only have shown up in live use. Each one is now handled in the design.

What we found	What we did about it
Out of the box, the platform quietly sends usage data to several outside companies.	Every one of these is switched off, and a firewall blocks anything else from leaving our servers. A network recording during testing will prove it.
By default, call recordings could be opened by anyone who knew where to look.	Recordings can now only be played from BongoHive's own office networks.
Callers could talk over the "you are speaking to an AI" notice and cut it short.	The notice is now played by the switchboard before the AI joins the call, so every caller hears all of it.
The speech engine would not start without an internet connection.	Models are now downloaded during planned maintenance, so Buzz runs fully offline day to day.
The 90-day clean-up would also have deleted the recorded AI notice itself.	The clean-up now tells call recordings apart from the system's own announcements.
The platform cannot pass a reason for a transfer to the staff phone.	Staff hear a short recorded introduction, optionally with the topic, before deciding to take the call.
Some numbers could cost money or reach the wrong place.	Buzz only dials real Zambian mobile and landline numbers, using the December 2024 national numbering plan. Everything else is refused before dialling.

The full technical detail is in the project plan (section 12) and the decisions log.

5 · NEXT STEPS

What happens **next**

The next phase is testing on real equipment. First we need a few decisions and some information from BongoHive.

Decisions and information needed

	What is needed	Why
1	**Server details** for the AI server and the main server: addresses, access, and the graphics card model. Blocks everything else.	Testing cannot start without them.
2	**Approval of the plan changes** listed in the decisions log (eight design choices made during the build).	So the build matches what BongoHive agrees to.
3	**Phone line**: a test account from the phone-line provider and their network details.	Needed for real test calls.
4	**Web address** for the staff page (for example voice.bongohive.co.zm) and how its security certificate is issued.	Staff log in here.
5	**Front-desk content**: programmes, coworking plans and prices, events, opening hours, and which team takes which transfers.	This is what Buzz will say.
6	**Buzz's voice**: staff choose one of four British English voices.	The voice callers hear.
7	**Data protection choices**: after 90 days, delete call records or anonymise them; the HubSpot consent field; where the n8n automation tool runs.	For the data protection impact assessment.
8	**Operations**: who receives alerts and how, the two people who keep the backup keys, which office networks may play recordings, and a few phone numbers set aside for testing.	Keeps the service safe to run.

5 · NEXT STEPS

How we will **test it**

Once the information above is in, testing runs in eight stages. Each stage builds on the one before, and ends with a clear pass or fail.

	Stage	Done when
1	Set up the AI server and measure how fast Buzz replies.	It runs offline and replies within the agreed time.
2	Set up the main platform and the staff web page.	Staff can log in; nothing leaves the network.
3	Connect the two and load the front-desk assistant.	Buzz answers correctly in a browser test call.
4	Connect the phone line and make 14 test calls.	Incoming, outgoing, transfers and refusals all behave as designed.
5	Test the safety rules from start to finish.	"Don't call me" blocks the number; no sensitive data reaches HubSpot.
6	Record all network traffic during testing.	Proof that no call data leaves BongoHive.
7	Check the dashboard and alerts, and restore a backup.	Alerts arrive; a backup restores cleanly.
8	Run the full set of test conversations, then 10 calls at once.	All required conversations pass; speed holds under load. Then the go-live decision.

The team's step-by-step testing plan is in **docs/TESTING.md** in the project repository.

REFERENCE

A few **terms explained**

Term	Meaning
AI language model	The part that understands what the caller said and decides how to reply. Buzz uses Qwen, an open model that runs on our own server.
Speech to text	Turns the caller's voice into written words (Whisper).
Text to speech	Turns Buzz's reply into a spoken voice (Kokoro).
Graphics card (GPU)	The specialised hardware that lets the AI reply quickly. Its memory size limits how many calls Buzz can handle at once.
Switchboard (Asterisk)	Open-source phone system software that connects Buzz to an ordinary phone line.
Conversation manager (Dograh)	Open-source software for designing and running AI voice conversations, set up here to work only with our own servers.
Opt-out / do-not-call list	The list of people who have asked not to be called. It is checked before every outgoing call.
Staging	A test copy of the system on real equipment, used to prove everything works before real callers use it.